# HW 7

## Question 1

Using the flights dataset from the package nycflights13, provide code to find:

1. All flights that had an arrival delay of two or more hours

```
flights |>
  filter(arr_delay >= 120)
```

```
# A tibble: 10,200 x 19
    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
 1  2013     1     1      811            630       101     1047            830
 2  2013     1     1      848           1835       853     1001           1950
 3  2013     1     1      957            733       144     1056            853
 4  2013     1     1     1114            900       134     1447           1222
 5  2013     1     1     1505           1310       115     1638           1431
 6  2013     1     1     1525           1340       105     1831           1626
 7  2013     1     1     1549           1445        64     1912           1656
 8  2013     1     1     1558           1359       119     1718           1515
 9  2013     1     1     1732           1630        62     2028           1825
10  2013     1     1     1803           1620       103     2008           1750
# i 10,190 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dttm>
```

2. All flights that flew to Houston (IAH or HOU)

```
flights |>
  filter(dest == 'IAH' | dest == 'HOU')
```

```
# A tibble: 9,313 x 19
    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
 1  2013     1     1      517            515         2      830            819
 2  2013     1     1      533            529         4      850            830
 3  2013     1     1      623            627        -4      933            932
 4  2013     1     1      728            732        -4     1041           1038
 5  2013     1     1      739            739         0     1104           1038
 6  2013     1     1      908            908         0     1228           1219
 7  2013     1     1     1028           1026         2     1350           1339
 8  2013     1     1     1044           1045        -1     1352           1351
 9  2013     1     1     1114            900       134     1447           1222
10  2013     1     1     1205           1200         5     1503           1505
# i 9,303 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dttm>
```

3. All flights that departed in summer (July, August, and September)

```r
flights |>
  filter(month == '7' | month == '8' | month == '9')
```

```
# A tibble: 86,326 x 19
    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
 1  2013     7     1        1           2029       212      236           2359
 2  2013     7     1        2           2359         3      344            344
 3  2013     7     1       29           2245       104      151              1
 4  2013     7     1       43           2130       193      322             14
 5  2013     7     1       44           2150       174      300            100
 6  2013     7     1       46           2051       235      304           2358
 7  2013     7     1       48           2001       287      308           2305
 8  2013     7     1       58           2155       183      335             43
 9  2013     7     1      100           2146       194      327             30
10  2013     7     1      100           2245       135      337            135
# i 86,316 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dttm>
```

4. All flights that departed between midnight and 6am (inclusive)

```r
flights |>
  filter(dep_time >= 0 & dep_time <= 600)
```

```
# A tibble: 9,344 x 19
    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
 1  2013     1     1      517            515         2      830            819
 2  2013     1     1      533            529         4      850            830
 3  2013     1     1      542            540         2      923            850
 4  2013     1     1      544            545        -1     1004           1022
 5  2013     1     1      554            600        -6      812            837
 6  2013     1     1      554            558        -4      740            728
 7  2013     1     1      555            600        -5      913            854
 8  2013     1     1      557            600        -3      709            723
 9  2013     1     1      557            600        -3      838            846
10  2013     1     1      558            600        -2      753            745
# i 9,334 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dttm>
```

5. All flights that arrived more than two hours late, but didn't leave late

```r
flights |>
  filter(arr_delay > 120 & dep_delay < 0)
```

```
# A tibble: 26 x 19
    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
 1  2013     1    27     1419           1420        -1     1754           1550
 2  2013    10     7     1357           1359        -2     1858           1654
 3  2013    10    16      657            700        -3     1258           1056
 4  2013    11     1      658            700        -2     1329           1015
 5  2013     3    18     1844           1847        -3       39           2219
 6  2013     4    17     1635           1640        -5     2049           1845
 7  2013     4    18      558            600        -2     1149            850
 8  2013     4    18      655            700        -5     1213            950
 9  2013     5    22     1827           1830        -3     2217           2010
10  2013     6     5     1604           1615       -11     2041           1840
# i 16 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
```

```
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dttm>
```

6. All flights that were delayed on the departure but made up some time in flight, so that their arrival delay wasn't as bad

```r
flights |>
  filter(dep_delay > 0 & arr_delay < dep_delay)
```

```
# A tibble: 83,728 x 19
    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
 1  2013     1     1      601            600         1      844            850
 2  2013     1     1      623            610        13      920            915
 3  2013     1     1      632            608        24      740            728
 4  2013     1     1      644            636         8      931            940
 5  2013     1     1      646            645         1      910            916
 6  2013     1     1      646            645         1     1023           1030
 7  2013     1     1      701            700         1     1123           1154
 8  2013     1     1      732            729         3     1041           1039
 9  2013     1     1      732            645        47     1011            941
10  2013     1     1      743            730        13     1107           1100
# i 83,718 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dttm>
```

### Question 2:

Write two different ways to use select() to extract the variables dep_time, sched_dep_time, and dep_delay from flights.

```r
flights |>
  select(dep_time:dep_delay)
```

```
# A tibble: 336,776 x 3
   dep_time sched_dep_time dep_delay
      <int>          <int>     <dbl>
 1      517            515         2
 2      533            529         4
```

```
 3      542            540         2
 4      544            545        -1
 5      554            600        -6
 6      554            558        -4
 7      555            600        -5
 8      557            600        -3
 9      557            600        -3
10      558            600        -2
# i 336,766 more rows
```

```
flights |>
  select(contains('dep'))
```

```
# A tibble: 336,776 x 3
   dep_time sched_dep_time dep_delay
      <int>          <int>     <dbl>
 1      517            515         2
 2      533            529         4
 3      542            540         2
 4      544            545        -1
 5      554            600        -6
 6      554            558        -4
 7      555            600        -5
 8      557            600        -3
 9      557            600        -3
10      558            600        -2
# i 336,766 more rows
```

### Question 3:

Write code to add the variable arrival, where arrival is a categorical variable that is equal to "ontime" when arr_delay <= 0 and "delayed" when arr_delay > 0.

```
flights |>
  mutate(arrival = if_else(arr_delay <= 0, 'ontime', 'delayed'))
```

```
# A tibble: 336,776 x 20
    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
 1  2013     1     1      517            515         2      830            819
```

```
 2  2013     1     1      533            529         4      850            830
 3  2013     1     1      542            540         2      923            850
 4  2013     1     1      544            545        -1     1004           1022
 5  2013     1     1      554            600        -6      812            837
 6  2013     1     1      554            558        -4      740            728
 7  2013     1     1      555            600        -5      913            854
 8  2013     1     1      557            600        -3      709            723
 9  2013     1     1      557            600        -3      838            846
10  2013     1     1      558            600        -2      753            745
# i 336,766 more rows
# i 12 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dttm>, arrival <chr>
```

### Question 4:

Currently dep_time and sched_dep_time are convenient to look at, but hard to compute with because they're not really continuous numbers. Use them to create a new variable for number of minutes since midnight. (Hint: use modular division)

```r
flights |>
  mutate(dep_time = (dep_time %/% 100) * 60 + dep_time %% 100,
         sched_dep_time = (dep_time %/% 100) * 60 + dep_time %% 100)
```

```
# A tibble: 336,776 x 19
    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>    <dbl>          <dbl>     <dbl>    <int>          <int>
 1  2013     1     1      317            197         2      830            819
 2  2013     1     1      333            213         4      850            830
 3  2013     1     1      342            222         2      923            850
 4  2013     1     1      344            224        -1     1004           1022
 5  2013     1     1      354            234        -6      812            837
 6  2013     1     1      354            234        -4      740            728
 7  2013     1     1      355            235        -5      913            854
 8  2013     1     1      357            237        -3      709            723
 9  2013     1     1      357            237        -3      838            846
10  2013     1     1      358            238        -2      753            745
# i 336,766 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dttm>
```

### Question 5:

Provide code to sort flights jointly by airline (carrier) and by longest to shortest departure delay within airline.

```
flights |>
  arrange(carrier, desc(dep_delay))
```

```
# A tibble: 336,776 x 19
    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
 1  2013     2    16      757           1930       747     1013           2149
 2  2013     7    24     1525            815       430     1808           1030
 3  2013    11    27     1503            815       408     1628            952
 4  2013     2    27     1529            845       404     1639           1015
 5  2013     7     7     2228           1559       389       NA           1828
 6  2013     6    25     1421            805       376     1602            950
 7  2013     1    25       15           1815       360      208           1958
 8  2013    12    14     1425            825       360     1604            938
 9  2013     9    12     2159           1600       359     2400           1818
10  2013     7    22     2216           1620       356      116           1853
# i 336,766 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dttm>
```

### Question 6:

Provide code to create a dataframe named avg_delay that has variables for each airline's average departure delay and average arrival delay.

```
avg_delay <- flights |>
  summarize(avg_dep_delay = mean(dep_delay, na.rm = T),
            avg_arr_delay = mean(arr_delay, na.rm = T),
            .by = carrier)
```

### Question 7:

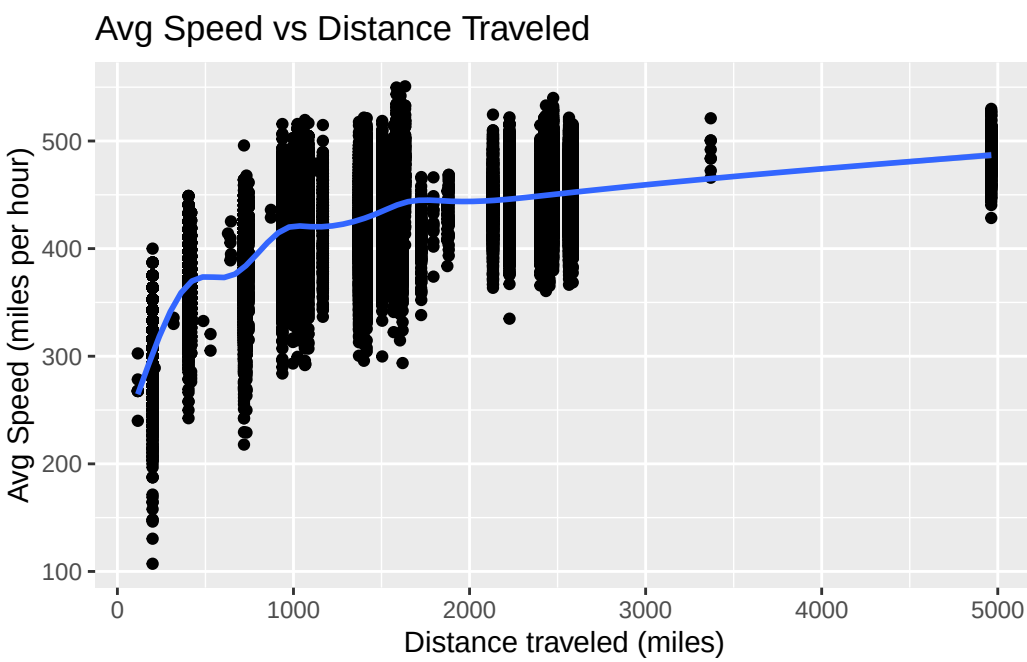
Create a scatterplot of average speed on the y axis (in miles per hour) by distance traveled on the x axis for all of United Airlines flights, and add a smooth trend to the plot.

```r
flights |>
  filter(carrier == 'UA') |>
  ggplot(aes(x = distance, y = (distance/(air_time/60)))) +
  geom_point() +
  geom_smooth() +
  labs(
    title = 'Avg Speed vs Distance Traveled',
    x = 'Distance traveled (miles)',
    y = 'Avg Speed (miles per hour)'
  )
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

```
Warning: Removed 883 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 883 rows containing missing values or values outside the scale range
(`geom_point()`).
```




Describe the relationship. Why do you think it looks the way it does?

The scatterplot shows a positive, nonlinear relationship between distance traveled and average speed. As the distance of a flight increases, the average speed generally increases as well, but the increase slows and begins to level off for longer distances. This pattern occurs because short flights spend a larger portion of their total time taking off and landing, when the plane is moving more slowly. In contrast, longer flights spend more time at cruising altitude, where planes travel at higher and relatively constant speeds. As a result, the average speed increases with distance until it approaches the typical cruising speed of the aircraft.

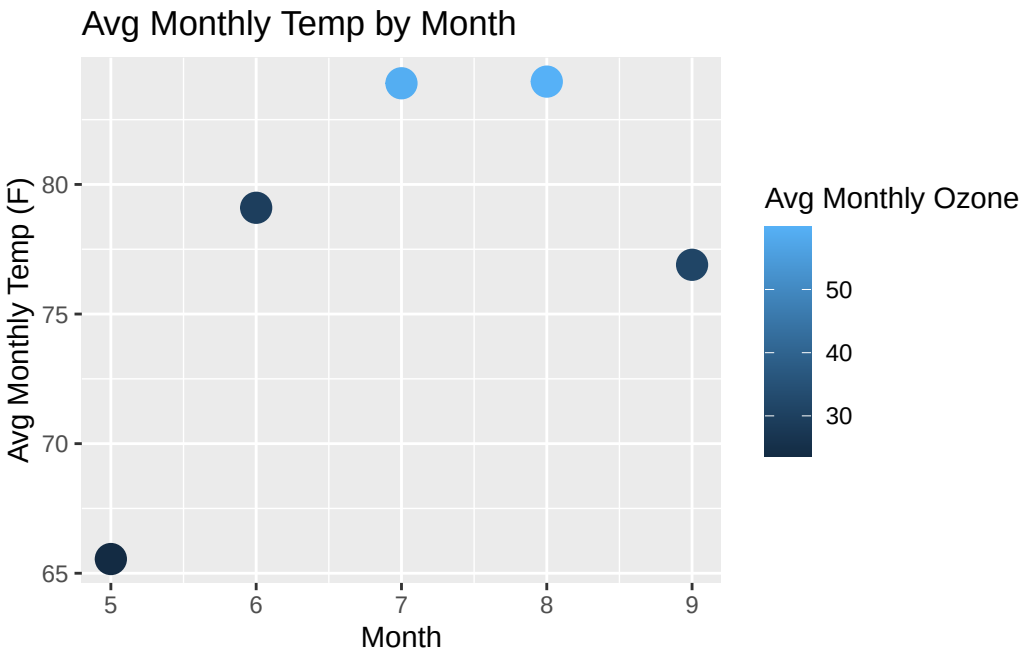
### Question 8:

Use the data set called "airquality" by using the command data("airquality")

```
data('airquality')
```

Using this dataset, create a scatterplot of average monthly temperature (y-axis) by month (x-axis), where the points are colored by the amount of average monthly Ozone.

```
airquality |>
  group_by(Month) |>
  summarize(avg_monthly_temp = mean(Temp, na.rm = T),
            avg_monthly_ozone = mean(Ozone, na.rm = T)) |>
  ggplot(aes(x = Month,
             y = avg_monthly_temp,
             color = avg_monthly_ozone)) +
  geom_point(size = 5) +
  labs(
    title = 'Avg Monthly Temp by Month',
    x = 'Month',
    y = 'Avg Monthly Temp (F)',
    color = 'Avg Monthly Ozone'
  )
```


Avg Monthly Temp by Month
Avg Monthly Temp (F)
80
75
70
65
5
6
7
8
9
Month
Avg Monthly Ozone
50
40
30